

JOIN in SQL

Join Operations in SQL Standard

- Inner Join
- Left Outer Join
- Right Outer Join
- Full Outer Join
- Equi Join
- Natural Join
- Cross Join

Inner Join

Employee table	
LastName	DepartmentID
Rafferty	31
Jones	33
Heisenberg	33
Robinson	34
Smith	34
Williams	NULL

```
SELECT employee.LastName, employee.DepartmentID, department.DepartmentName
FROM employee INNER JOIN department ON
    employee.DepartmentID = department.DepartmentID;
```

Employee. LastName	Employee. DepartmentID	Department. DepartmentName
Robinson	34	Clerical
Jones	33	Engineering
Smith	34	Clerical
Heisenberg	33	Engineering
Rafferty	31	Sales

Department table	
DepartmentID	DepartmentName
31	Sales
33	Engineering
34	Clerical
35	Marketing

```
SELECT E.LastName, E.DepartmentID, D.DepartmentName
FROM employee E, department D
WHERE E.DepartmentID = D.DepartmentID;
```

Equi Join

- An **equi-join** is a specific type of comparator-based join, that uses only equality comparisons in the join-predicate. Using other comparison operators (such as <) disqualifies a join as an equi-join.

```
SELECT *  
FROM employee JOIN department ON employee.DepartmentID = department.DepartmentID;
```

```
SELECT *  
FROM employee, department  
WHERE employee.DepartmentID = department.DepartmentID;
```

```
SELECT *  
FROM employee INNER JOIN department USING (DepartmentID);
```

Natural Join

- The natural join is a special case of equi-join. Natural join ($\bowtie$) is a [binary operator](#) that is written as $(R \bowtie S)$ where R and S are [relations](#). The result of the natural join is the set of all combinations of [tuples](#) in R and S that are equal on their common attribute names.

SELECT * FROM employee NATURAL JOIN department;

DepartmentID	Employee.LastName	Department.DepartmentName
34	Smith	Clerical
33	Jones	Engineering
34	Robinson	Clerical
33	Heisenberg	Engineering
31	Rafferty	Sales

Employee table	
LastName	DepartmentID
Rafferty	31
Jones	33
Heisenberg	33
Robinson	34
Smith	34
Williams	NULL

Left Outer Join

```

SELECT *
FROM employee LEFT OUTER JOIN department ON
    employee.DepartmentID = department.DepartmentID;

```

Department table	
DepartmentID	DepartmentName
31	Sales
33	Engineering
34	Clerical
35	Marketing

Employee. LastName	Employee. DepartmentID	Department. DepartmentName	Department. DepartmentID
Rafferty	31	Sales	31
Jones	33	Engineering	33
Heisenberg	33	Engineering	33
Robinson	34	Clerical	34
Smith	34	Clerical	34
Williams	NULL	NULL	NULL

Employee table	
LastName	DepartmentID
Rafferty	31
Jones	33
Heisenberg	33
Robinson	34
Smith	34
Williams	NULL

Right Outer Join

```

SELECT *
FROM employee RIGHT OUTER JOIN department ON
      employee.DepartmentID = department.DepartmentID;

```

Department table	
DepartmentID	DepartmentName
31	Sales
33	Engineering
34	Clerical
35	Marketing

Employee. LastName	Employee. DepartmentID	Department. DepartmentName	Department. DepartmentID
Rafferty	31	Sales	31
Jones	33	Engineering	33
Heisenberg	33	Engineering	33
Robinson	34	Clerical	34
Smith	34	Clerical	34
NULL	NULL	Marketing	35

Full Outer Join

```
SELECT *  
FROM employee FULL OUTER JOIN department ON  
employee.DepartmentID = department.DepartmentID;
```

Employee table	
LastName	DepartmentID
Rafferty	31
Jones	33
Heisenberg	33
Robinson	34
Smith	34
Williams	NULL

Department table	
DepartmentID	DepartmentName
31	Sales
33	Engineering
34	Clerical
35	Marketing

Employee. LastName	Employee. DepartmentID	Department. DepartmentName	Department. DepartmentID
Rafferty	31	Sales	31
Jones	33	Engineering	33
Heisenberg	33	Engineering	33
Robinson	34	Clerical	34
Smith	34	Clerical	34
Williams	NULL	NULL	NULL
NULL	NULL	Marketing	35

Cross Join

- CROSS JOIN returns the [Cartesian product](#) of rows from tables in the join. In other words, it will produce rows which combine each row from the first table with each row from the second table. For the below example, the output table will have $6 \times 4 = 24$ rows.

SELECT * FROM employee **CROSS JOIN** department;

Employee.LastName	Employee.DepartmentID	Department.Department Name	Department.DepartmentID
Rafferty	31	Sales	31
Jones	33	Sales	31
Heisenberg	33	Sales	31
Smith	34	Sales	31
Robinson	34	Sales	31
Williams	NULL	Sales	31
Rafferty	31	Engineering	33
Jones	33	Engineering	33
...	...	...	...